Problems In Group Theory

Basu Dev Karki

Contents

Chapter 1

Basic Group Theory

1.1 Introduction

Problem 1.1.1. *Determine which of the following binary operations are associative:*

1. *the operation on* $\mathbb{Z}$ *defined by* $a * b = a - b$
2. *the operation on* $\mathbb{R}$ *defined by* $a * b = a + b + ab$
3. *the operation on* $\mathbb{Q}$ *defined by* $a * b = \frac{a+b}{5}$
4. *the operation on* $\mathbb{Z} \times \mathbb{Z}$ *defined by* $(a, b) * (c, d) = (ad + bc, bd)$
5. *the operation on* $\mathbb{Q} - \{0\}$ *defined by* $a * b = \frac{a}{b}$

Proof. This exercise is pretty trivial as we just have to compute $a * (b * c) = (a * b) * c$. For 1., $a * (b * c) = a * (b - c) = a - (b - c) = a - b + c$ but $(a * b) * c = (a - b) * c = (a - b) - c = a - b - c$. Rest of them are pretty similar. □

Problem 1.1.2. *Prove that addition of residue classes in* $\mathbb{Z}/n\mathbb{Z}$ *is associative.*

Proof. By definition of residue addition,

$$[a] + [b] = [a + b]$$

Therefore, we get the following.

$$\begin{aligned} [a] + ([b] + [c]) &= [a] + [b + c] \\ &= [a + (b + c)] \\ &= [(a + b) + c] \\ &= [a + b] + [c] \\ &= ([a] + [b]) + [c] \end{aligned}$$

□

Problem 1.1.3. *Prove that multiplication of residue classes in* $\mathbb{Z}/n\mathbb{Z}$ *is associative.*

Proof. By definition of residue multiplication,

$$[a] \cdot [b] = [a \cdot b]$$

Therefore, we get the following.

$$\begin{aligned} [a] \cdot ([b] \cdot [c]) &= [a] \cdot [b \cdot c] \\ &= [a \cdot (b \cdot c)] \\ &= [(a \cdot b) \cdot c] \\ &= [a \cdot b] \cdot [c] \\ &= ([a] \cdot [b]) \cdot [c] \end{aligned}$$

□

Problem 1.1.4. *Prove for all $n > 1$ that $\mathbb{Z}/n\mathbb{Z}$ is not a group under multiplication of residue classes.*

Proof. There is no multiplicative inverse for $[a]$ with $\gcd(a, n) > 1$. The proof for it pretty trivial, if there was a inverse, it would mean that

$$a \cdot b \equiv 1 \pmod{n} \implies 0 \equiv 1 \pmod{\gcd(a, n)}$$

That's a contradiction. □

Problem 1.1.5. *Determine which of the following sets are groups under addition:*

1. *the set of rational numbers (including $0 = 0/1$) in lowest terms whose denominators are odd*
2. *the set of rational numbers (including $0 = 0/1$) in lowest terms whose denominators are even*
3. *the set of rational numbers of absolute value < 1*
4. *the set of rational numbers of absolute value ≥ 1 together with 0*
5. *the set of rational numbers with denominators equal to 1 or 2*
6. *the set of rational numbers with denominators equal to 1, 2 or 3.*

Proof. Below are the answers:

1. Group.
2. Not a group. Because it is not closed. (Eg: $5/2 + 7/2 = 6$)
3. Not a group. Because again it is not closed. (Eg : $0.99 + 0.98 = 1.97$)
4. Not a group. Because again it is not closed. (Eg : $4 + (-3.1) = 0.9$)
5. Group.
6. Not a group. Because its not closed. (Eg : $1/2 + 1/3 = 5/6$)

□

Problem 1.1.6. *Let $G = \{x \in \mathbb{R} \mid 0 \leq x < 1\}$ and for $x, y \in G$ let $x * y$ be the fractional part of $x + y$ (i.e., $x * y = x + y - \lfloor x + y \rfloor$ where $\lfloor a \rfloor$ is the greatest integer less than or equal to a). Prove that $*$ is a well defined binary operation on G and that G is an abelian group under $*$ (called the real numbers mod 1).*

Proof. It is well defined as,

$$x_1 = x_2, y_1 = y_2 \implies x_1 * y_1 = x_1 + y_1 - \lfloor x_1 + y_1 \rfloor$$
$$\implies x_1 + y_1 - \lfloor x_1 + y_1 \rfloor = x_2 + y_2 - \lfloor x_2 + y_2 \rfloor = x_2 * y_2$$

Also, its closed as $x * y = \{x + y\}$ and $0 \leq x + y < 1$. One can check that inverses and identity exists. For associativity,

$$\begin{aligned} x * (y * z) &= x * (y + z - \lfloor y + z \rfloor) \\ &= \{x + y + z - \lfloor y + z \rfloor\} \\ &= \{x + y + z\} \\ &= \{x + y - \lfloor x + y \rfloor + z\} \\ &= (x * y) * z \end{aligned}$$

□

Problem 1.1.7. *Let $G = \{z \in \mathbb{C} \mid z^n = 1 \text{ for some } n \in \mathbb{Z}^+\}$.*

1. *Prove that G is a group under multiplication (called the group of roots of unity in $\mathbb{C}$).*
2. *Prove that G is not a group under addition.*

Proof. For G under multiplication, we know that for all $z_1, z_2, z_3 \in G$ we have

$$z_1 \cdot (z_2 \cdot z_3) = (z_1 \cdot z_2) \cdot z_3$$

Also, it is closed as for all $z_1, z_2 \in G$ we have $z_1^n = 1$ and $z_2^k = 1$ so

$$(z_1 \cdot z_2)^{\text{lcm}(n,k)} = 1$$

Now, for inverse of a $z \in G$ just take $1/z$. As $(1/z)^n = 1/1 = 1$ so $1/z \in G$. Also, $z \cdot 1/z = 1$.
For G under addition, if it was a group then, $1 \in G$ as $1^1 = 1 \in G \implies 1 + 1 = 2 \in G$ but there does not exist any integer n such that $2^n = 1$. □

Problem 1.1.8. *Let $G = \{a + b\sqrt{2} \in \mathbb{R} \mid a, b \in \mathbb{Q}\}$.*

1. *Prove that G is a group under addition.*
2. *Prove that the nonzero elements of G are a group under multiplication.*

Proof. We know reals are associative so the elements of G are associative under addition. The inverse for $a + b\sqrt{2}$ is $(-a) + (-b)\sqrt{2}$. It is also closed as,

$$a + b\sqrt{2} + c + d\sqrt{2} = (a + c) + (b + d)\sqrt{2}$$

The identity is going to be $0 + 0 \cdot \sqrt{2} = 0$.
We know reals are associative so the elements of G are associative under multiplication. The inverse for $a + b\sqrt{2}$ is $\dfrac{a}{a^2 - 2b^2} - \dfrac{b}{a^2 - 2b^2}\sqrt{2}$. The identity is $1 + 0 \cdot \sqrt{2} = 1$.
And its closed under multiplication as

$$(a + b\sqrt{2})(c + d\sqrt{2}) = (ac + 2bd) + (ad + bc)\sqrt{2}$$

□